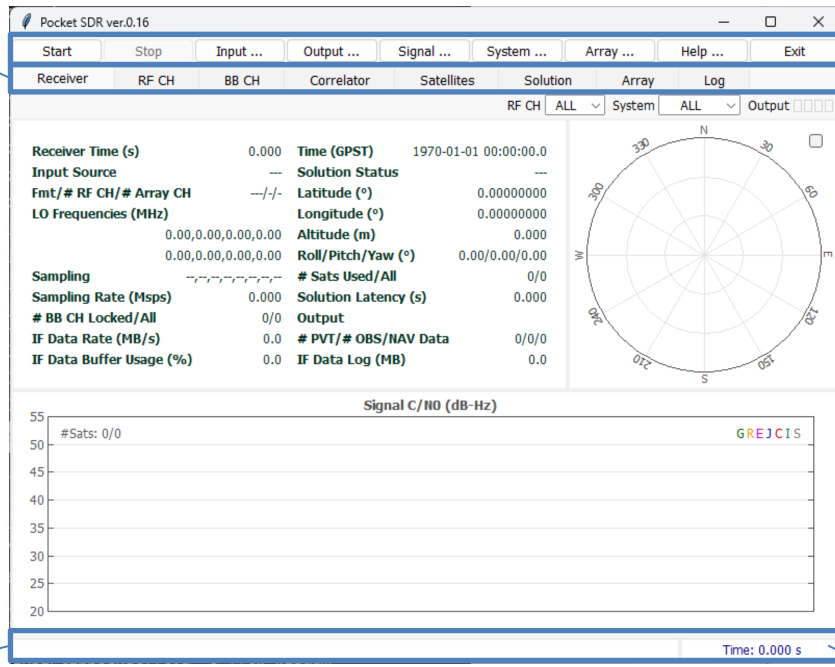
pocket_sdr.py help

ver. 0.16 2026-06-01

Initial Window

Page Tabs:
Tabs to switch current page



Command Buttons

Start	: Start the receiver
Stop	: Stop the receiver
Input...	: Show Input Options dialog
Output...	: Show Output Options dialog
Signal...	: Show Signal Options dialog
System...	: Show System Options dialog
Array...	: Show Array Options dialog
Help	: Show Help dialog
Exit	: Exit the receiver

Status Bar:
Status info or error message for the receiver

Receiver Time:
Elapsed time since the receiver started (s)

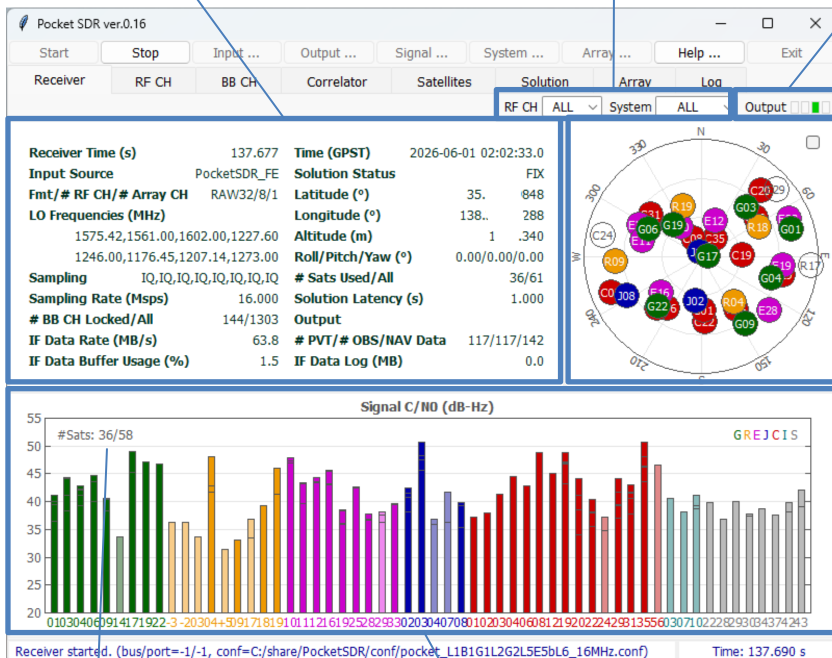
Receiver Page

Receiver Time:
Elapsed time since the receiver started
Input Source:
IF data source (RF_Frontend or IF_Data)
Fmt/# RF CH/# Array CH:
IF data format *3 and number of CHs
LO Frequencies (MHz):
Local oscillator frequencies for each RF CH (MHz)
Sampling:
IF data sampling type for each RF CH (I or IQ)
Sampling Rate (Msps):
IF data sampling rate
 # BB CH Locked/All:
Number of BB CHs tracked and all
IF Data Rate (MB/s):
IF data transfer rate
IF Data Buffer Usage (%):
Receiver internal IF data buffer usage rate
Time (GPST):
PVT Solution time in GPST
Solution Status:
PVT Solution status (- or FIX)
Latitude, Longitude and Altitude:
PVT Solution latitude/ longitude (deg) and ellipsoidal height (m)
Roll/Pitch/Yaw (deg):
Attitude solution by antenna array
 # Sats Used / All:
Number of used satellites for PVT and all satellites with OBS data
Solution Latency (s):
PVT Solution latency
 # PVT/# OBS/NAV Data:
Total count of PVT solutions, OBS and NAV data
IF Data Log (MB):
Recorded size of IF Data log

Receiver Status Panel

RF CH and System Selection for Skyplot and C/N0 Plot

Output Status Indicators
for PVT solution, OBS/NAV data, receiver log and IF data log
Orange: waiting for connection
Green: data OK
Light-Green: data active
Red: error
White: no output



Satellite Positions in Skyplot *1 *2

Number of used satellites and all satellites with OBS data

Signal C/N0 (dB-Hz) Plot for Each Satellite and Signals (stacked) *1 *2

* 1 GNSS System Color:
Green: GPS
Orange: GLONASS
Magenta: Galileo
Blue: QZSS
Red: BeiDou
Blue-Green: NavIC
Grey: SBAS
White or light-color: unused satellites for solution

*2 Satellite ID complies with RINEX convention
*3 INT8 (int8 I-sampling), INT8X2 (interleaved-int8 IQ-sampling), RAW8, RAW16, RAW32 (Pocket SDR FE), CS8 or CS16 (SoapySDR devices)

RF CH Page

Link to GNSS Signal Band Table

Sampling Rate

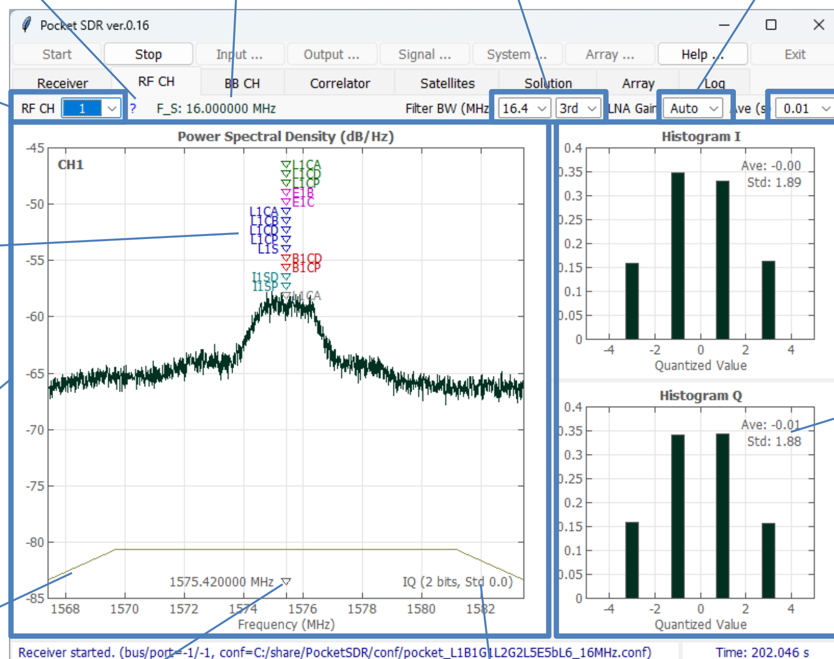
IF Filter Bandwidth and order Selection *1 *2

LNA Gain Selection *1

RF CH Selection:
ALL, 1, 2, ..., 16, 1-4 or 5-8 *3

Averaging Time Selection for Power Spectral Density and Histogram

Carrer-Frequency Marks for each GNSS Signal



Power Spectral Density Plot (Spectral Analyzer) for each RF CH

Average and Standard Deviation of Sample Values

IF Filter Bandwidth

Histogram for IF Data Sample Values (I-sample and Q-sample)

Local Oscillator Frequency

Sampling Type (I or IQ) and number of sample bits, Standard-deviation of Sample Value

*1 Only valid for RF_Frontend Input, *2 Only valid for IQ-sampling
*3 RF CH 9, 10, ..., 16 are beamforming CHs by antenna array

BB CH Page

RF CH Selection: ALL, 1, 2, ..., 16

GNSS System Selection: ALL, GPS, GLONASS, ...

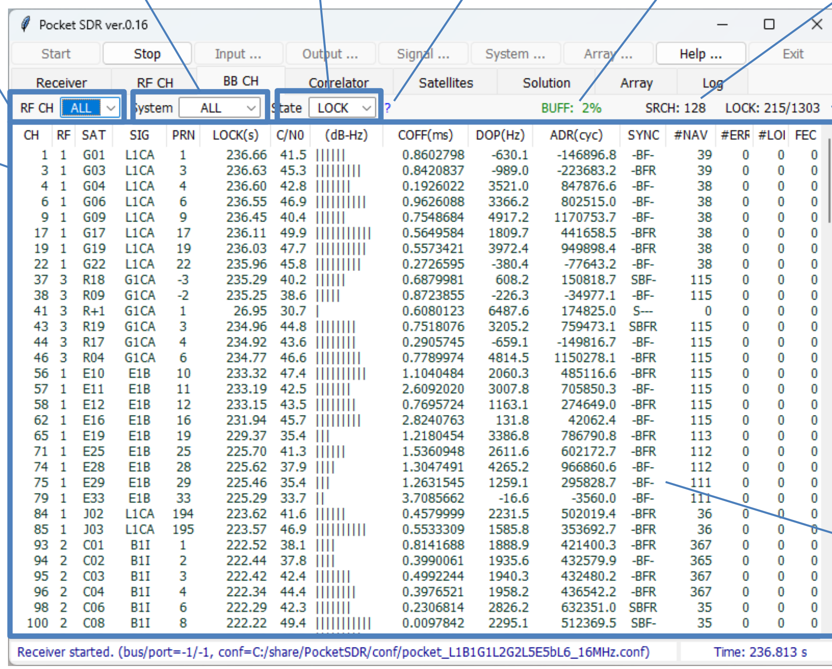
BB CH State Selection: LOCK or ALL

Link to GNSS Signal Table

Internal IF Data Buffer Usage Rate in %

BB CH No Searching Signal

Number of Locked and All BB CHs



CH	RF	SAT	SIG	PRN	LOCK(s)	C/N0	(dB-Hz)	COFF(ms)	DOP(Hz)	ADR(cyc)	SYNC	#NAV	#ERR	#LOL	FEC
1	1	G01	L1CA	1	236.66	41.5		0.8602798	-630.1	-146896.8	-BF-	39	0	0	0
3	1	G03	L1CA	3	236.63	45.3		0.8420837	-989.0	-223683.2	-BFR	39	0	0	0
4	1	G04	L1CA	4	236.60	42.8		0.1926022	3521.0	847876.6	-BF-	38	0	0	0
6	1	G06	L1CA	6	236.55	46.9		0.9626088	3366.2	802515.0	-BF-	38	0	0	0
9	1	G09	L1CA	9	236.45	40.4		0.7548684	4917.2	1170753.7	-BF-	38	0	0	0
17	1	G17	L1CA	17	236.11	49.9		0.5649584	1809.7	441658.5	-BFR	38	0	0	0
19	1	G19	L1CA	19	236.03	47.7		0.5573421	3972.4	949898.4	-BFR	38	0	0	0
22	1	G22	L1CA	22	235.96	45.8		0.2726595	-380.4	-77643.2	-BF-	38	0	0	0
37	3	R18	G1CA	-3	235.29	40.2		0.6879981	608.2	150818.7	SBF-	115	0	0	0
38	3	R09	G1CA	-2	235.25	38.6		0.8723855	-226.3	-34977.1	-BF-	115	0	0	0
41	3	R+1	G1CA	1	26.95	30.7		0.6080123	6487.6	174825.0	S---	0	0	0	0
43	3	R19	G1CA	3	234.96	44.8		0.7518076	3205.2	759473.1	SBFR	115	0	0	0
44	3	R17	G1CA	4	234.92	43.6		0.2905745	-659.1	-149816.7	-BF-	115	0	0	0
46	3	R04	G1CA	6	234.77	46.6		0.7789974	4814.5	1150278.1	-BFR	115	0	0	0
56	1	E10	E1B	10	233.32	47.4		1.1040484	2060.3	485116.6	-BFR	115	0	0	0
57	1	E11	E1B	11	233.19	42.5		2.6092020	3007.8	705850.3	-BF-	115	0	0	0
58	1	E12	E1B	12	233.15	43.5		0.7695724	1163.1	274649.0	-BFR	115	0	0	0
62	1	E16	E1B	16	231.94	45.7		2.8240763	131.8	42062.4	-BF-	115	0	0	0
65	1	E19	E1B	19	229.37	35.4		1.2180454	3386.8	786790.8	-BFR	113	0	0	0
71	1	E25	E1B	25	225.70	41.3		1.5360948	2611.6	602172.7	-BFR	112	0	0	0
74	1	E28	E1B	28	225.62	37.9		1.3047491	4265.2	966860.6	-BF-	112	0	0	0
75	1	E29	E1B	29	225.46	35.4		1.2631545	1259.1	295828.7	-BF-	111	0	0	0
79	1	E33	E1B	33	225.29	33.7		3.7085662	-16.6	-3560.0	-BF-	111	0	0	0
84	1	J02	L1CA	194	223.62	41.6		0.4579999	2231.5	502019.4	-BFR	36	0	0	0
85	1	J03	L1CA	195	223.57	46.9		0.5533309	1585.8	353692.7	-BFR	36	0	0	0
93	2	C01	B1I	1	222.52	38.1		0.8141688	1888.9	421400.3	-BFR	367	0	0	0
94	2	C02	B1I	2	222.44	37.8		0.3990061	1935.6	432579.9	-BF-	365	0	0	0
95	2	C03	B1I	3	222.42	42.4		0.4992244	1940.3	432480.2	-BFR	367	0	0	0
96	2	C04	B1I	4	222.34	44.4		0.3976521	1958.2	436542.2	-BFR	367	0	0	0
98	2	C06	B1I	6	222.29	42.3		0.2306814	2826.2	632351.0	SBFR	35	0	0	0
100	2	C08	B1I	8	222.22	49.4		0.0097842	2295.1	512369.5	SBF-	35	0	0	0

Status for each BB CH:

- CH : BB CH no
- RF : Assigned RF CH no for BB CH
- SAT : Satellite ID
- SIG : Signal ID
- PRN : PRN number or FCN for GLONASS FDMA signal
- LOCK : Continuous lock time (s)
- C/N0 : Signal C/N0 (dB-Hz)
- (dB-Hz) : Signal C/N0 Bar (20-50 dB-Hz)
- COFF : Code offset (ms)
- DOP : Doppler frequency (Hz)
- ADR : Accumulated Doppler range (cycle)
- SYNC : Synchronization status
- #NAV : Decoded subframe or message count
- #ERR : NAV data error count
- #LOL : Loss-of-lock count
- FEC : Number of error bits corrected by FEC (-1: correction unable)

Synchronization Status:

- S : Secondary code sync
- B : NAV symbol (bit) sync
- F : Subframe or message sync
- R : Polarity of subframe or message sync (R: reversed)

Correlator Page

BB CH Selection *1: 1, 2, 3, …

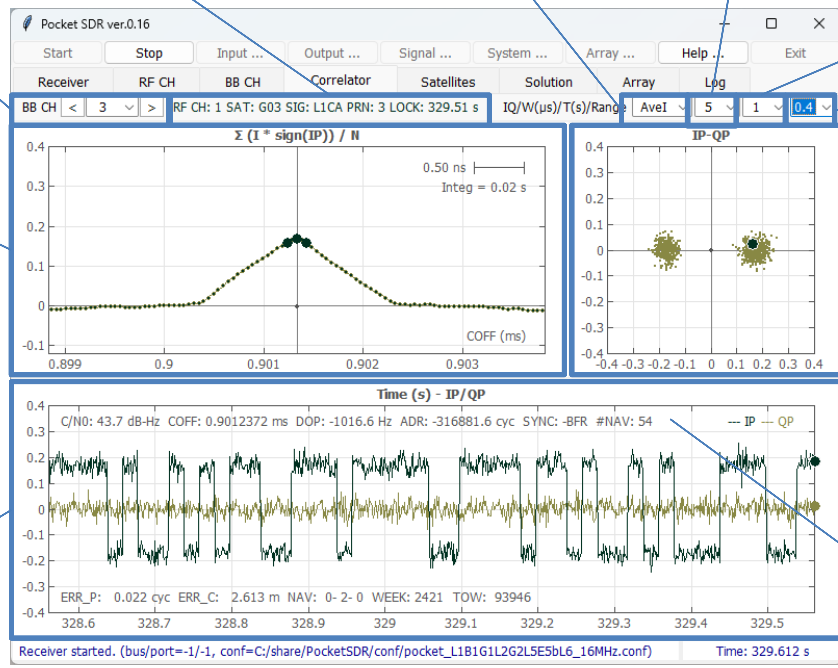
RF CH, Satellite ID, Signal ID PRN No. and Lock Time of BB CH

Correlation Plot Type Selection

X-axis Range Selection for Correlation Plot

Time Span Selection for Correlation History or Constellation Plot

Y-axis Range Selection



Correlation Plot

X-axis: code offset (ms)
Y-axis: correlations
(I * sign(IP), sqrt($I^2 + Q^2$) or sqrt(sum($I^2 + Q^2$))) *2
Large dots: correlator positions for VE, E, P, L and VL *3

Correlation History Plot

X-axis: receiver time (s)
Y-axis: IP or QP-correlation

Constellation Plot

X-axis: IP-correlation
Y-axis: QP-correlation
Large dot: current correlation

C/N0, code offset, Doppler Frequency, accumulated Doppler range, synchronization status, decoded subframe or message count

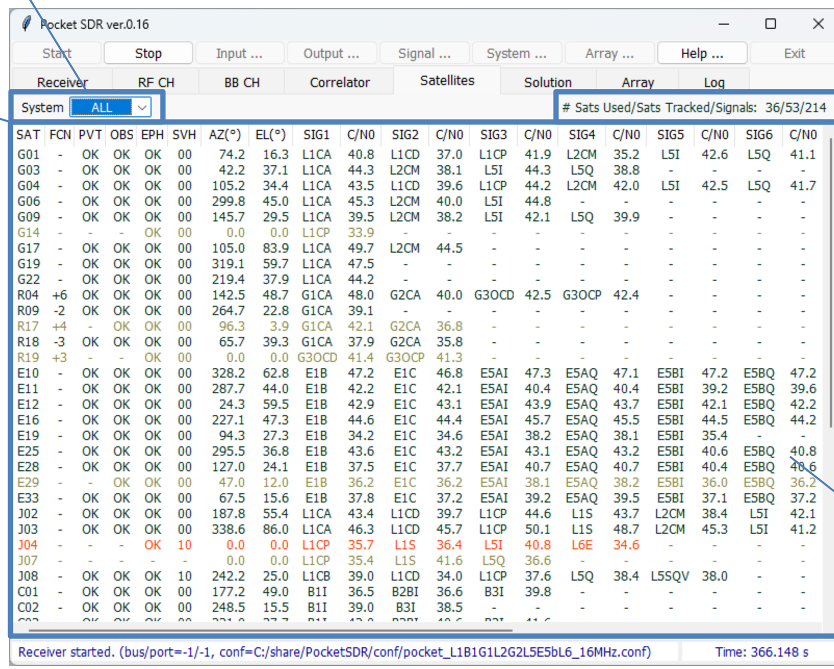
*1 BB CH can be selected by clicking BB CH Status row in BB CH Page
*2 Integration time is based on non-coherent integration time for DLL
*3 VE and VL are only shown in case of BOC-modulation and Bump-Jump enabled

Satellites Page

System Selection

Status for each Satellite:

SAT : Satellite ID
FCN : Frequency channel no (FCN) for GLONASS FDMA signals
PVT : Satellite used in PVT solution or not
OBS : Satellite L1 OBS data availability
EPH : Satellite ephemeris availability
SVH : Satellite health flag in ephemeris (HEX)
AZ : Azimuth angle (deg)
EL : Elevation angle (deg)
SIG1 : Signal 1 ID
C/N0 : C/N0 of signal 1 (dB-Hz)
SIG2 : Signal 2 ID
C/N0 : C/N0 for signal 2 (dB-Hz)
SIG3 : Signal 3 ID
C/N0 : C/N0 for signal 3 (dB-Hz)
... ...



SAT	FCN	PVT	OBS	EPH	SVH	AZ(°)	EL(°)	SIG1	C/N0	SIG2	C/N0	SIG3	C/N0	SIG4	C/N0	SIG5	C/N0	SIG6	C/N0
G01	-	OK	OK	OK	00	74.2	16.3	L1CA	40.8	L1CD	37.0	L1CP	41.9	L2CM	35.2	L5I	42.6	L5Q	41.1
G03	-	OK	OK	OK	00	42.2	37.1	L1CA	44.3	L2CM	38.1	L5I	44.3	L5Q	38.8	-	-	-	-
G04	-	OK	OK	OK	00	105.2	34.4	L1CA	43.5	L1CD	39.6	L1CP	44.2	L2CM	42.0	L5I	42.5	L5Q	41.7
G06	-	OK	OK	OK	00	299.8	45.0	L1CA	45.3	L2CM	40.0	L5I	44.8	-	-	-	-	-	-
G09	-	OK	OK	OK	00	145.7	29.5	L1CA	39.5	L2CM	38.2	L5I	42.1	L5Q	39.9	-	-	-	-
G14	-	-	-	OK	00	0.0	0.0	L1CP	33.9	-	-	-	-	-	-	-	-	-	-
G17	-	OK	OK	OK	00	105.0	83.9	L1CA	49.7	L2CM	44.5	-	-	-	-	-	-	-	-
G19	-	OK	OK	OK	00	319.1	59.7	L1CA	47.5	-	-	-	-	-	-	-	-	-	-
G22	-	OK	OK	OK	00	219.4	37.9	L1CA	44.2	-	-	-	-	-	-	-	-	-	-
R04	+6	OK	OK	OK	00	142.5	48.7	G1CA	48.0	G2CA	40.0	G3OCD	42.5	G3OCP	42.4	-	-	-	-
R09	-2	OK	OK	OK	00	264.7	22.8	G1CA	39.1	-	-	-	-	-	-	-	-	-	-
R17	+4	-	OK	OK	00	96.3	3.9	G1CA	42.1	G2CA	36.8	-	-	-	-	-	-	-	-
R18	-3	OK	OK	OK	00	65.7	39.3	G1CA	37.9	G2CA	35.8	-	-	-	-	-	-	-	-
R19	+3	-	-	OK	00	0.0	0.0	G3OCD	41.4	G3OCP	41.3	-	-	-	-	-	-	-	-
E10	-	OK	OK	OK	00	328.2	62.8	E1B	47.2	E1C	46.8	E5AI	47.3	E5AQ	47.1	E5BI	47.2	E5BQ	47.2
E11	-	OK	OK	OK	00	287.7	44.0	E1B	42.2	E1C	42.1	E5AI	40.4	E5AQ	40.4	E5BI	39.2	E5BQ	39.6
E12	-	OK	OK	OK	00	24.3	59.5	E1B	42.9	E1C	43.1	E5AI	43.9	E5AQ	43.7	E5BI	42.1	E5BQ	42.2
E16	-	OK	OK	OK	00	227.1	47.3	E1B	44.6	E1C	44.4	E5AI	45.7	E5AQ	45.5	E5BI	44.5	E5BQ	44.2
E19	-	OK	OK	OK	00	94.3	27.3	E1B	34.2	E1C	34.6	E5AI	38.2	E5AQ	38.1	E5BI	35.4	-	-
E25	-	OK	OK	OK	00	295.5	36.8	E1B	43.6	E1C	43.2	E5AI	43.1	E5AQ	43.2	E5BI	40.6	E5BQ	40.8
E28	-	OK	OK	OK	00	127.0	24.1	E1B	37.5	E1C	37.7	E5AI	40.7	E5AQ	40.7	E5BI	40.4	E5BQ	40.6
E29	-	-	OK	OK	00	47.0	12.0	E1B	36.2	E1C	36.2	E5AI	38.1	E5AQ	38.2	E5BI	36.0	E5BQ	36.2
E33	-	OK	OK	OK	00	67.5	15.6	E1B	37.8	E1C	37.2	E5AI	39.2	E5AQ	39.5	E5BI	37.1	E5BQ	37.2
J02	-	OK	OK	OK	00	187.8	55.4	L1CA	43.4	L1CD	39.7	L1CP	44.6	L1S	43.7	L2CM	38.4	L5I	42.1
J03	-	OK	OK	OK	00	338.6	86.0	L1CA	46.3	L1CD	45.7	L1CP	50.1	L1S	48.7	L2CM	45.3	L5I	41.2
J04	-	-	-	OK	10	0.0	0.0	L1CP	35.7	L1S	36.4	L5I	40.8	L6E	34.6	-	-	-	-
J07	-	-	-	-	-	0.0	0.0	L1CP	35.4	L1S	41.6	L5Q	36.6	-	-	-	-	-	-
J08	-	OK	OK	OK	10	242.2	25.0	L1CB	39.0	L1CD	34.0	L1CP	37.6	L5Q	38.4	L5SQV	38.0	-	-
C01	-	OK	OK	OK	00	177.2	49.0	B1I	36.5	B2BI	36.6	B3I	39.8	-	-	-	-	-	-
C02	-	OK	OK	OK	00	248.5	15.5	B1I	39.0	B3I	38.5	-	-	-	-	-	-	-	-

Number of satellites used for PVT solution, number of satellites tracked, number of all tracked signals

Red: unhealthy satellite
Grey: unused satellite for PVT solution

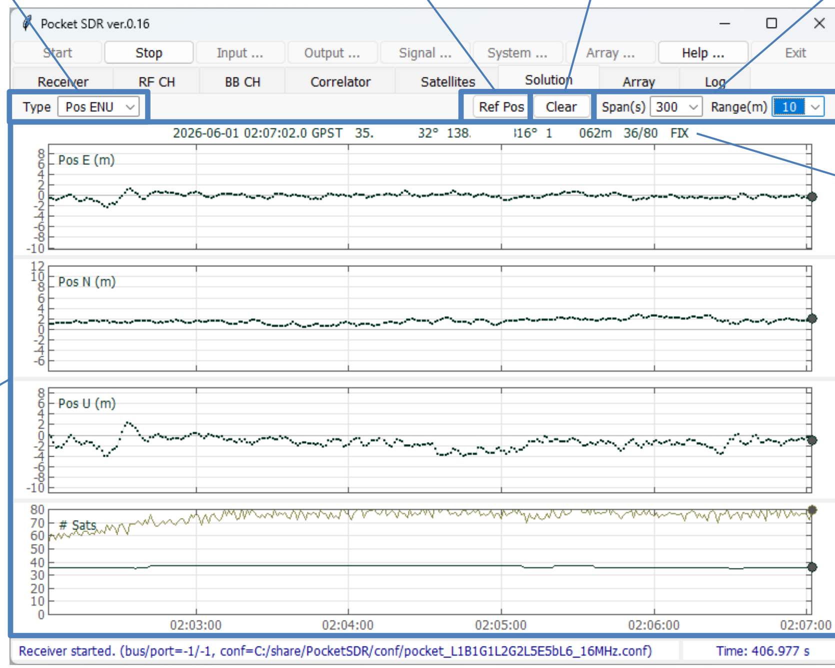
Solution Page

PVT Solution Plot Selection:
Pos ENU or Pos Horiz

Show Ref Pos Dialog Button

PVT Solution Clear Button

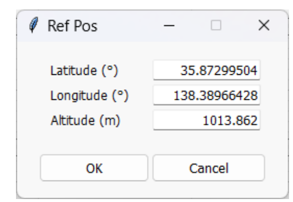
Time Span and Y-axis Range Selections



Current Solution and Number of Used/All Satellites

PVT Solution Plot:
Pos E, N, U: position errors east-west, north-south, up-down *1,
Sats: Number of used/all satellites
or
Position error horizontal *1
(east-west and north-west, horizontal scatter plot)

Ref Pos Dialog



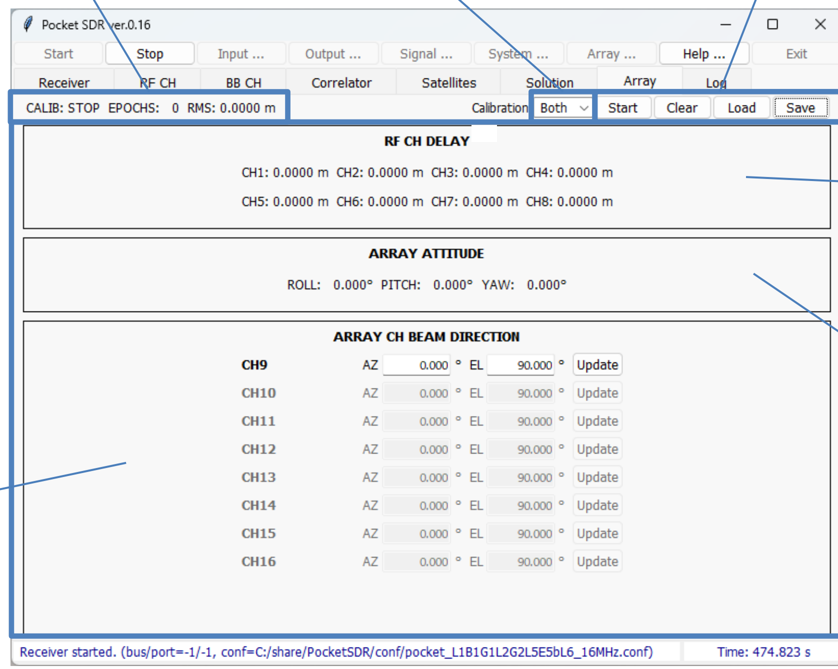
*1 The reference position is automatically selected from the first epoch solution.
It can be modified by using Ref Pos Dialog

Array Page

Array Calibration Status:
STOP or RUN, Total epochs for calibration and RMS residuals

Array Calibration Targets:
Both: Dealy and Attiude
Delay: Delay only
Att: Attitude only

Array Calibration Operation Buttons:
Start or Stop, Clear, Load and Save



RF CH DELAY:
RF CH delay calibration results referenced to RF CH1

ARRAY ATTITUDE:
Array attitude calibration results (ROLL, PITCH, YAW angles)

ARRAY CH BEAM DIRECTION:
Beam direction control for each Array CH. Set AZ (azimuth) and EL (elevation) angles and push Update Button.

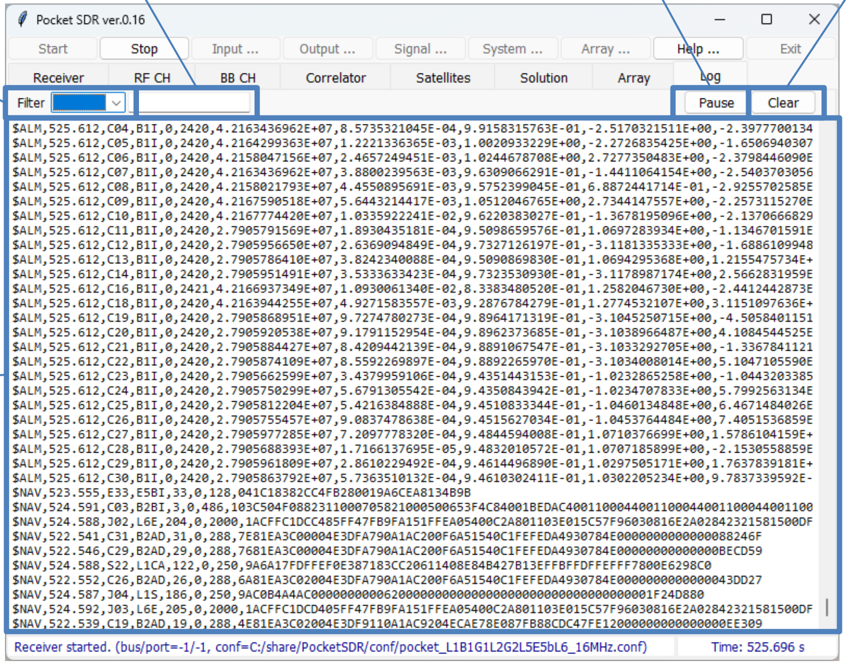
Log Page

Additional Receiver Log Filter:
filter string, like G01 or L1CA, separated by spaces, | indicates OR condition (e.g., G10|G02|G03)

Receiver Log Pause Button

Receiver Log Clear Button

Receiver Log Type Selection:
Blank: all log



Receiver Logs *1 *2

*1 Refer p.16 for the receiver log formats

*2 To copy the logs to Clip-board, push Pause button, select the logs and push Ctrl-C of the keyboard

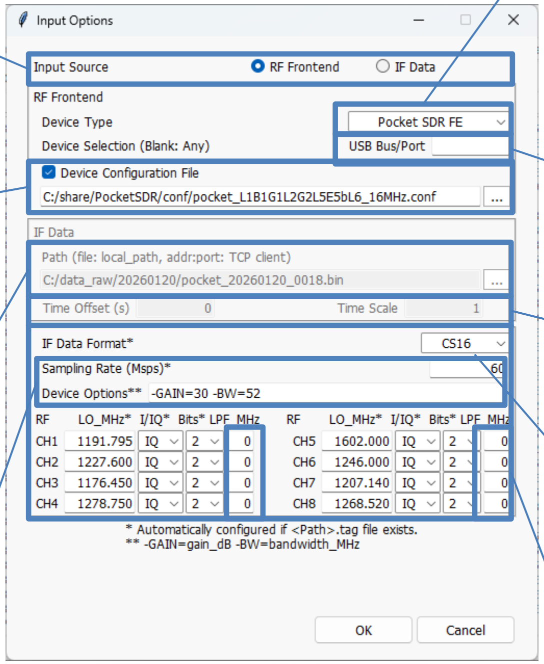
Input Options Dialog

Input Source Selection:
RF Frontend or IF Data

Device Configuration File:
RF Frontend configuration settings for Pocket SDR FE (MAX2771 register) to be loaded at the receiver started. Check Device Configuration File and fill the file path. If unchecked, the current configuration for the device is used. Ignored for other devices.

IF Data File *1:
IF data file path, if IF Data selected as the input source. The IF data can be captured by pocket_dump or as IF data log of the receiver with tag files.

Sampling Rate and Device Options for Soapy SDR Devices:
-GAIN=gain (LNA gain),
-BW=bw (Filter Bandwidth),
Options are separated by spaces.



RF Frontend Type Selection:
Pocket SDR FE: Pocket SDR FE 2CH, 4CH, 8CH
USRP (uhd): USRP via Soapy SDR
LimeSDR (lime): LimeSDR via Soapy
LimeSDR (limesuiteng): LimeSDR via newer driver with Soapy
BladeRF (bladrf): bladeRF via Soapy
RTL-SDR (rtlsdr): RTL-SDR via Soapy
Plute-SDR (plutosdr): PlutoSDR via Soapy
Airspy (airspy): Airspy via Soapy

USB Bus/Port:
To select Pocket SDR FE in case of multiple devices connected. If blank, the device is automatically selected.

Time Offset and Scale:
Start time offset and time scale to playback the IF Data.

IF Data File Property *1:
IF Data Format, Sampling Rate, LO frequency, Sampling Type (I or IQ), Number of sample bits for each RF CH. If the tag file exist for the IF data file, these are automatically configured according to the tag file.

Input LPF Settings:
Set two-sided bandwidth (MHz) of LPF for each RF CH. 0 indicates disabled.

*1 Refer to p.15 for the IF data file formats

Output Options Dialog

PVT Solution Output Path:

PVT Solution (NMEA 0183 GNRMC, GNGGA, GNGSA, GxGSV) path *1.

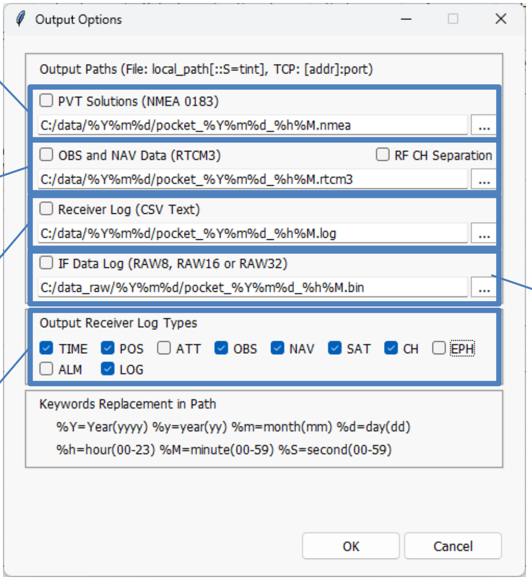
OBS and NAV Data Output Path:

OBS and NAV Data (RTCM 3.4 MSM (MT1077, 1087, 1097, 1117, 1127, 1137 and 1107) with extension and EPH (MT1019, 1020, 1045, 1046, 1044, 1042 and 1041) path *1 *2

Receiver Log Output Path:

Receiver Log Path *1 *3

Receiver Log Type Selections *3



IF Data Log Output Path:

IF Data Log Path *1 *4, It is only valid for RF Frontend selected as the input source

*1 The output path should be one of the followings.

(1) <local path>: A local file path <local path>. It can contain time keywords replaced by the log start time. Optionally, it can include the file swapping order "::S=<tint>" by the specified interval <tint> (H).

(2) :<port>: TCP server with the port number <port>. The TCP server waits for and accepts the external TCP connection at the port and output data

(3) <addr>:<port>: TCP client with the address <addr> and the port number <port>. The TCP client connects the external TCP server and output data

*2 The output RTCM 3.4 MSM (with extensions) and EPH can be converted to RINEX OBS and NAV by CONVBIN utility included in the Pocket SDR package

*3 Refer to p.16 for the receiver log format. The receiver log can be plotted by the Log Viewer pocket_plot.py in the Pocket SDR package

*4 Refer p.15 for the IF data log format. The output format is automatically selected according to the RF Frontend (RAW8, RAW16, RAW32, CS8 or CS16). The tag file is also generated.

Signal Options Dialog

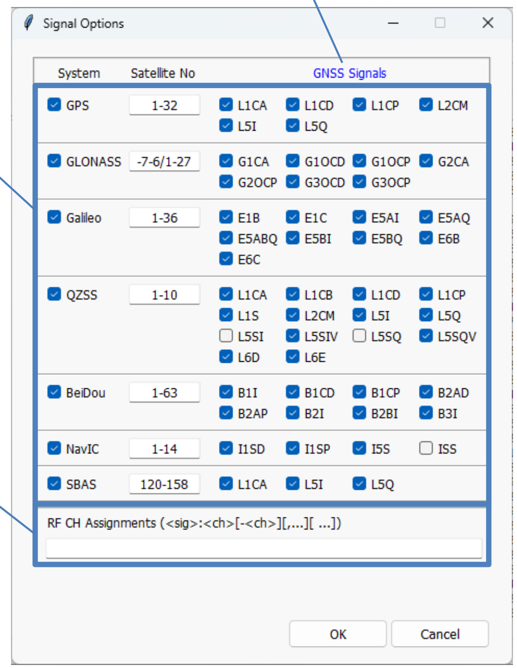
GNSS System and Signal Selections:

System: GNSS System,
Satellite No: Satellite number or number range (n-m) separated by , [*1]
GNSS Signals: Signal IDs to be tracked

RF CH Assignment Options:

RF CH is automatically assigned for signals. To assign signals to specific RF CH(s) manually, use the option as the format <sig>:<ch_list> separated by spaces, where, <sig> is signal ID and <ch_list> is RF CH number list like 1,3,5-7. Additionally, SRCH:<ch_list> can be used to specify signal search CH(s) to shorten the acquisition time.

Link to Detailed GNSS Signal Table



Pocket SDR Signal ID Table

System	Signal	Signal ID
GPS	L1C/A	L1CA
	L1C-D	L1CD
	L1C-P	L1CP
	L2C-M	L2CM
	L5-I	L5I
	L5-Q	L5Q
GLONASS	L1C/A	G1CA
	L1OCd	G1OCD
	L1OCp	G1OCP
	L2C/A	G2CA
	L2OCp	G2OCP
	L3OCd	G3OCD
	L3OCp	G3OCP
Galileo	E1-B	E1B
	E1-C	E1C
	E5a-I	E5AI
	E5a-Q	E5AQ
	E5AltBOC	E5ABQ
	E5b-I	E5BI
	E5b-Q	E5BQ
	E6-B	E6B
	E6-C	E6C
QZSS	L1C/A	L1CA
	L1C/B	L1CB
	L1C-D	L1CD
	L1C-P	L1CP
	L1S	L1S
	L2C-M	L2CM
	L5-I	L5I
	L5Q	L5Q
	L5S-I	L5SI
		L5SIV[*2]
	L5S-Q	L5SQ
		L5SQV[*2]
	L6D	L6D
	L6E	L6E

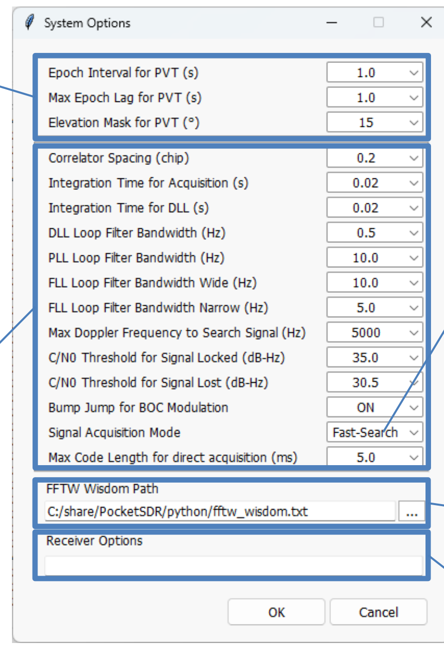
System	Signal	Signal ID
BeiDou	B1I	B1I
	B1C-D	B1CD
	B1C-P	B1CP
	B2a-D	B2AD
	B2a-P	B2AP
	B2I	B2I
	B2b-I	B2BI
	B3I	B3I
NavIC	L1-SPS-D	I1SD
	L1-SPS-P	I1SP
	L5-SPS	I5S
	S-SPS	ISS[*3]
SBAS	L1C/A	L1CA
	L5-I	L5I
	L5-Q	L5Q

*1 For GPS, Galileo, BeiDou, NavIC and SBAS, these satellite numbers are the same as the signal PRN numbers. For GLONASS, use the frequency channel number (FCN) for FDMA signals and satellite slot number for CDMA signals instead separated by /. For QZSS, which has complicated satellite numbering, use SV IDs (1-5, 8, 9) or non-standard code numbers (6, 10) instead as the satellite numbers.
*2 QZSS L5S verification mode signals
*3 Pocket SDR FE currently does not support S-band signals

System Options Dialog

PVT Options Selection:
Epoch Interval (s), Max Epoch Lag (s) and Elevation angle Mask (deg)

Signal Acquisition and Tracking Options Selection:
Correlation Spacing (chip), Non-coherent Integration Time for Acquisition (s), Non-coherent Integration Time for DLL (s), Loop filter bandwidth for DLL, PLL, FLL (wide-band) and FLL (narrow-band) (Hz), Max Doppler Frequency search range (Hz), C/N0 thresholds for signal-lock or signal-lost decision (dB-Hz), Bump-Jump for BOC modulation enabled or disabled, Max code length for direct acquisition (ms).



Signal Acquisition Mode Selection:
Full-Search: All PRN signals are searched. Fast-Search: Only signals of visible satellites are searched for fast acquisition. The visibility is estimated with CPU time, last or current FIX position, almanac or ephemeris of each satellites. The Doppler also is decided by last or current refence oscillator freq. offset. So, if FE changed, use Full-Search.

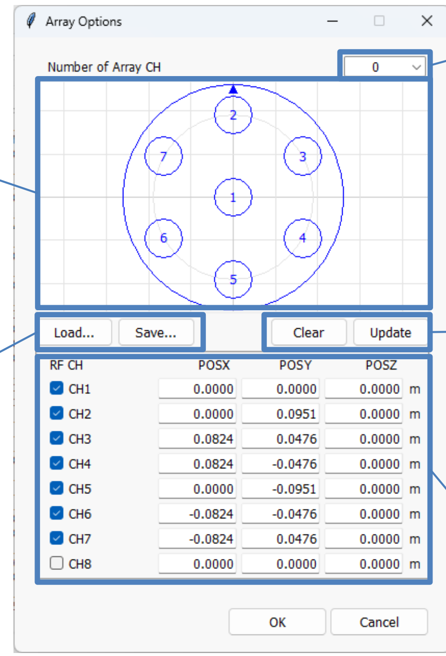
FFTW Wisdom Path:
FFTW wisdom file path to shorten the signal acquisition time. The file can be generated by the utility fftw_wisdom in the Pocket SDR package.

Receiver Specific Options

Array Options Dialog

Number of Array CH enabled.

Array Element Horizontal Plot (Antenna Geometry)

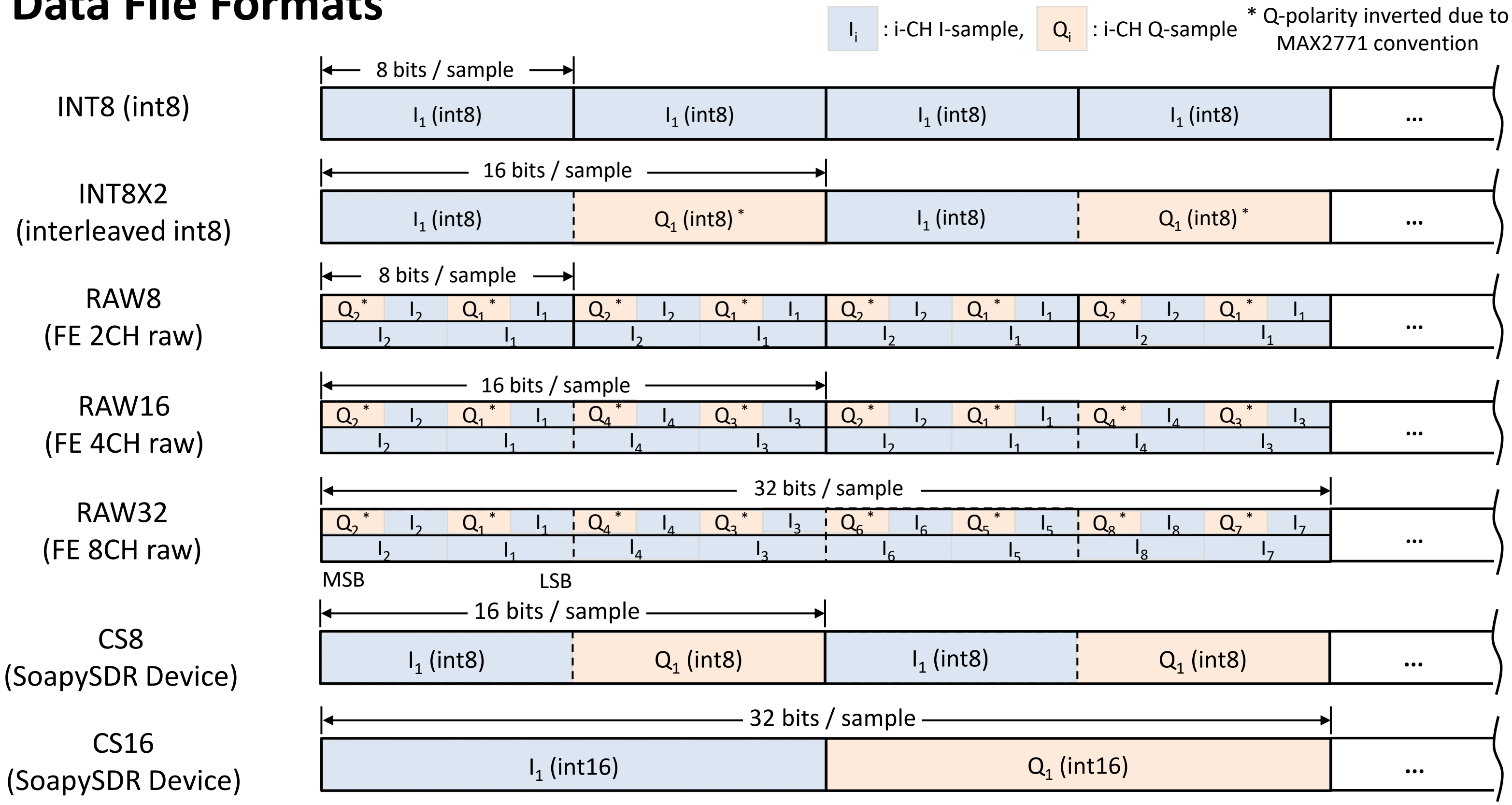


Antenna Geometry Clear and Update Buttons

Antenna Geometry Load and Save Buttons

Enable/Disable and Positions X, Y, Z (m) in body-frame for each Antenna Element.

IF Data File Formats



2 bits IQ-sampling (RAW8/16/32)

bits	00	01	10	11
I	+1	+3	-1	-3
Q *	-1	-3	+1	+3

2 bits I-sampling (RAW8/16/32)

bits	**00	**01	**10	**11
I	+1	+3	-1	-3

3 bits I-sampling (RAW8/16/32)

bits	0000	1000	0001	1001	0010	1010	0011	1011
I	+1	+3	+5	+7	-1	-3	-5	-7

IF Data Tag File (<path>.tag)

```
PROG = Pocket SDR
TIME = 2024/11/01 20:48:00.776     (Start Time GPST)
FMT  = RAW16                       (Format)
F_S  = 24                          (Sampling Freq. MHz)
F_LO = 1568,1237.8,1176.45,1278.75 (LO Freq. MHz)
IQ   = 2,2,2,2                     (Sampling, 1:I, 2:IQ)
BITS = 2,2,2,2                     (Sampling Bits)
```

Receiver Logs

```
$TIME,time,year,mon,day,hour,min,sec,tsys (time info)
$OBS,time,year,month,day,hour,min,sec,sat,code,cn0,pr,cp,dop,lli,fcn (OBS data)
$NAV,time,sat,sig,nerr,size,data (raw NAV data)
$POS,time,year,month,day,hour,min,sec,lat,lon,hgt,Q,ns,stdn,stde,stdu,dtr (position solution)
$SAT,time,year,month,day,hour,min,sec,sat,pvt,obs,cn0,az,el,res (satellite info)
$EPH,time,sat,sig,IODE,IODC,SVA,SVH,Toe,Toc,Ttr,A,e,i0,OMEGA0,omega,M0,delta-n,OMEGAdot,Idot,Crc,Crs,Cuc,Cus,Cic,Cis,Toes,Fit,Af0,Af1,Af2,
  TGD,code,flag (GPS/Galileo/QZSS/BeiDou/NavIC decoded ephemeris)
$EPH,time,sat,sig,tb,fcn,SVH,SVA,age,Toe,Tof,pos-x,pos-y,pos-z,vel-x,vel-y,vel-z,acc-x,acc-y,acc-z,tau-n,gamma-n,delta-tau-n (GLONASS
  decoded ephemeris)
$ALM,time,sat,sig,svh,week,A,e,i0,OMG0,omg,M0,OMGd,toas,f0,f1 (GPS,Galileo,QZSS,BeiDou decoded alamanac)
$ALM,time,sat,sig,svh,taun,lambda,di,eps,omg,tlambda,dT,dTd,frq (GLONASS decoded alamanac)
$CH,time,ch,rfch,sat,sig,prn,lock,cn0,coff,dop,adr,ssync,bsync,fsync,rev,srev,err_phas,err_code,tow_v,tow,week,type,nnav,nerr,nlol,nfec
  (receiver channel info)
$LOG,time,message (error, warning, general info message)
```

```
$TIME
time  receiver time (s)
year  year (2000-2099)
mon   month (1-12)
day   day (1-31)
hour  hour (0-23)
min   minute (0-59)
sec   second (0.000-59.999)
tsys  time system (GPST or UTC)

$OBS
time  receiver time (s)
year,month,day
      obs data day (GPST)
hour,min,sec
      obs data time (GPST)
sat   satellite ID
code  RINEX obs code
cn0   C/N0 (dB-Hz)
pr    pseudorange (m)
cp    carrier phase (cyc)
dop   Doppler frequency (Hz)
lli   loss-of-lock indicator
fcn   frequency channel number
      for GLONASS FDMA signals

$NAV
time  receiver time (s)
sat   satellite ID
sig   signal ID
nerr  number of corrected error
      bits
size  raw NAV data size (bits)
data  raw NAV data HEX dump

$POS
time  receiver time (s)
year,month,day  solution day (GPST)
hour,min,sec  solution time (GPST)
lat   solution latitude
      (deg,+:north,-:south)
lon   solution longitude
      (deg,+:east,-:west)
hgt   solution ellipsoidal height
      (m)
Q     quality flag (=5: single)
ns    number of valid satellites
stdn  solution standard-dev north
      (m)
stde  solution standard-dev east
      (m)
stdu  solution standard-dev up (m)
dtr   receiver clock bias (s)

$SAT
time  receiver time (s)
year,month,day  solution day (GPST)
hour,min,sec  solution time (GPST)
sat   satellite ID
pvt   PVT status (0:not used,1:used)
obs   L1 obs data status
      (0:unavailable,1:available)
cn0   L1 signal C/N0 (dB-Hz)
az    azimuth angle (deg)
el    elavation angle (deg)
res   L1 pseudorange residual (m)

$EPH
time  receiver time (s)
sat   satellite ID
sig   signal ID
...   decoded ephemeris parameters

$ALM
time  receiver time (s)
sat   satellite ID
sig   signal ID
...   decoded almanac parameters

$LOG
time  receiver time (s)
message  error, warning or general info message

$CH
time  receiver time (s)
ch    receiver channel number
rfch  RF channel number
sat   satellite ID
sig   signal ID
prn   PRN number
lock  lock time (s)
cn0   C/N0 (dB-Hz)
coff  code offset (ms)
dop   Doppler frequency (Hz)
adr   accumlated Doppler range (cyc)
ssync secondary code sync flag(0:async,1:sync)
bsync symbol/bit sync flag (0:async,1:sync)
fsync frame sync flag (0:async,1:sync)
rev   primary code polarity (0:normal,1:reversed)
srev  secondary code polarity (0:normal,1:reversed)
err_phas  phase error (cyc)
err_code  code error (10^-6 s)
tow_v  tow valid flag (0:invalid,1:valid,
       2:ambiguity unresolved)
tow   time of week (ms)
week  week number (week)
type  navigation subframe or message type
nnav  navigation subframe/message count
nerr  error subframe/message count
nlol  loss-of-lock count
nfec  number of error corrected (bits)
```